Feedback From Last Weekend

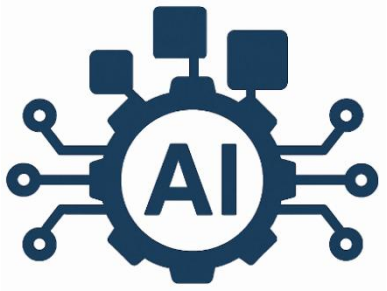
tired

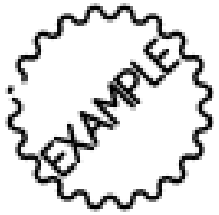
end earlier

fewer words per slide

Schedule For This Weekend

Date		Topics	DDL
Nov 15	AM	• Achieving Operational Excellence and Customer Intimacy (Laudon Ch9)	Proposal due: **Nov 15 at 23:59**
	PM	• Ethical and Social Issues in Information Systems (Laudon Ch4)	
Nov 16	AM	• Artificial Intelligence (Laudon Ch11)	
	PM	• Enhancing Decision Making with DA and BI (Laudon Ch12)	





温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W03
Strategic Mgt Global Info Sys

Achieving Operational Excellence and Customer Intimacy: Enterprise Applications

Dawei Chen

15 Nov 2025

Kick-Off Discussion

What **information systems** have you used or seen in an organization?

How did they help improve operations or strengthen customer relationships?

Learning Objectives

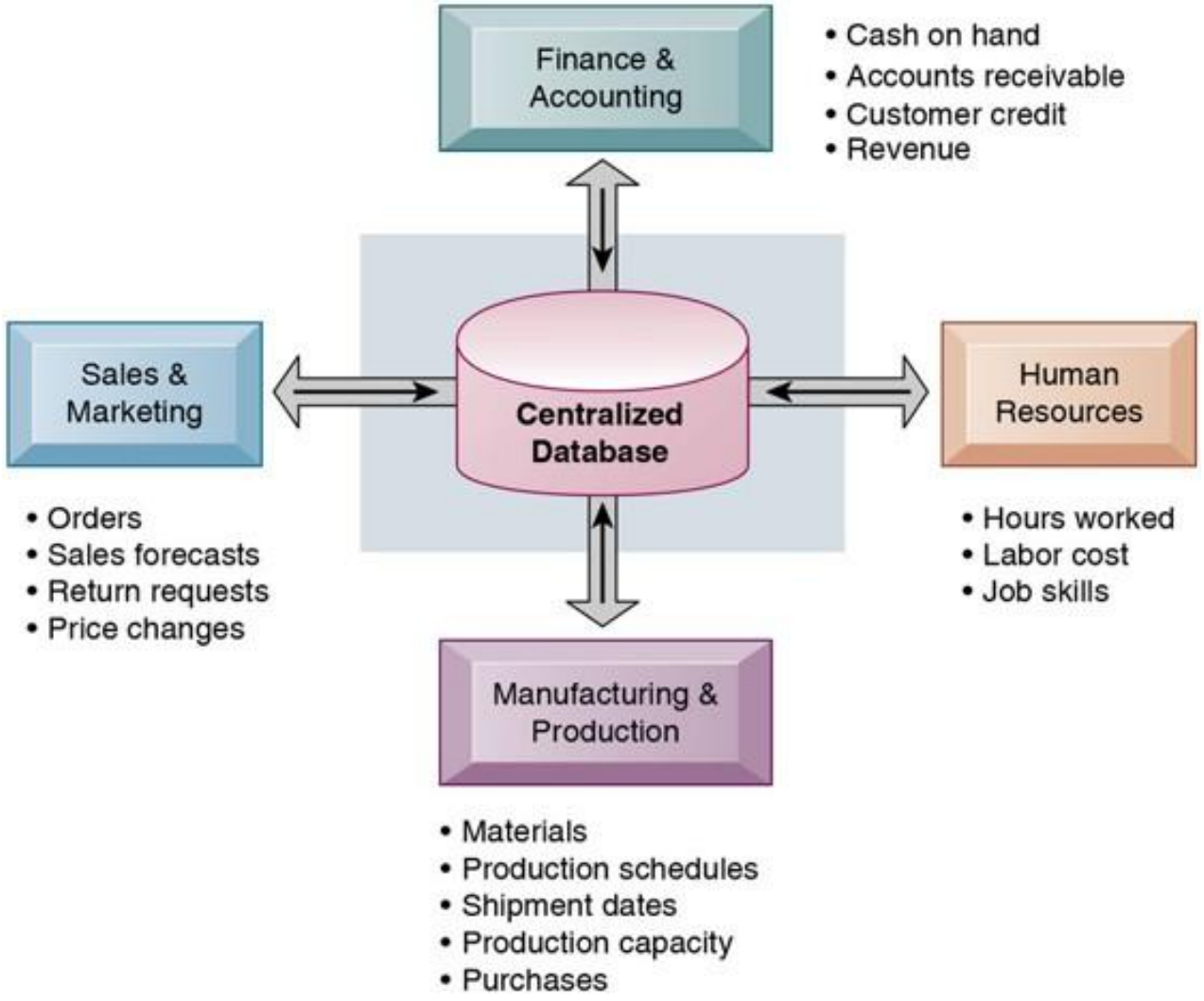
- Understand enterprise resource planning (ERP) systems
- Understand supply chain management (SCM) systems
- Describe global supply chain management challenges
- Understand customer relationship management (CRM) systems
- Discuss enterprise application challenges
- Describe how enterprise applications are using AI

ERP Systems – Definition

- Also known as enterprise resource planning (ERP) systems
- Based on a suite of integrated software modules and a common central database
- Collects data from many divisions of firm for use in nearly all of firm's internal business activities
- Information entered in one process is immediately available for other processes

Why is it valuable?

ERP Systems – Definition



Leading ERP software vendors:

SAP, Oracle, Microsoft, Salesforce, Blue Yonder

Example: [Oracle Fusion Cloud HCM]

ERP Systems – Implementation

1) Select Functions & Map Processes
   - Choose which ERP functions (finance, supply chain, HR, etc.) to deploy.
   - Map existing business processes to the system's predefined workflows.
2) Configure the System
   - Use vendor configuration tables to align ERP settings with business needs
     → e.g., track revenue by product line, region, or channel.
3) Avoid Excessive Customization
   - Customization can degrade performance and weaken integration.
   - Best results come from adapting processes to the ERP's best practices.

ERP Systems – Business Value

- Increased operational efficiency
- Provide firm-wide information to support decision making
- Enable rapid responses to customer requests for information or products
- Include analytical tools to evaluate overall organizational performance and improve decision-making

ERP Systems – AI Applications

Capability	Description
Digital assistant	Use AI-powered conversational chatbots to respond to customer and employee inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.
Predictive analytics	Anticipating equipment failures, recommending preventative maintenance schedules and procedures.
Anomaly detection	Analyzing ERP financial reports to identify irregularities in data entries or unusual transactions.
Improve product development and innovation	Suggest new product ideas, optimize product features and assortment.
Financial forecasting	Generating plausible financial scenarios based on potential economic, policy, or market changes.
Optimize workflows	AI can assist with almost every function of a modern supply chain, including planning, inventory and warehouse management, transaction processing, transportation, monitoring, and inspection.

ERP Systems – AI Applications

What's New and Next for Oracle AI in Fusion Applications: Oracle AI World 2025

Example: Use AI to Simplify Supplier Management in Oracle Procurement (Demo)

ERP Systems – Quiz

Which of the following statements about the ERP systems is **not true**?

A. Enterprise systems improve management decision making by providing more accurate sales and production forecasts.

B. ERP systems are built around thousands of predefined business processes that reflect industry best practices.

C. Configuration tables enable a company to tailor a particular aspect of an ERP system to the way a company does business.

D. To reap the maximum benefits from ERP software, companies should customize it to conform to their existing business processes.

SCM Systems – Supply Chain

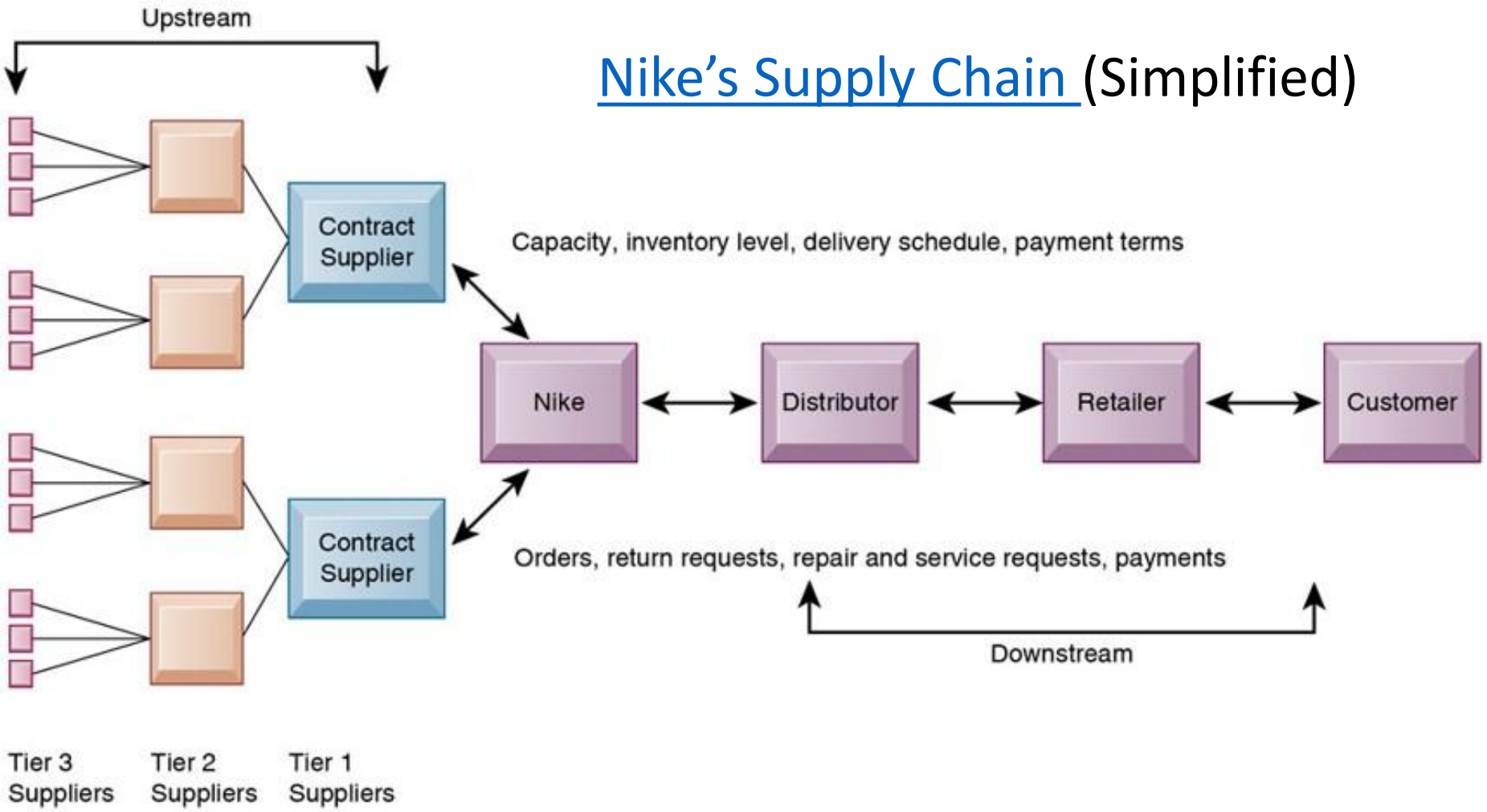
- A firm's supply chain is a **network** of organizations and business processes that procure raw materials, transform these materials into intermediate and finished products, and distribute the finished products to customers.

- Materials, information, and payments flow through the supply chain in **both directions**.

Oracle Fusion Cloud Supply Chain and Manufacturing

SCM Systems – Supply Chain

Nike's Supply Chain (Simplified)



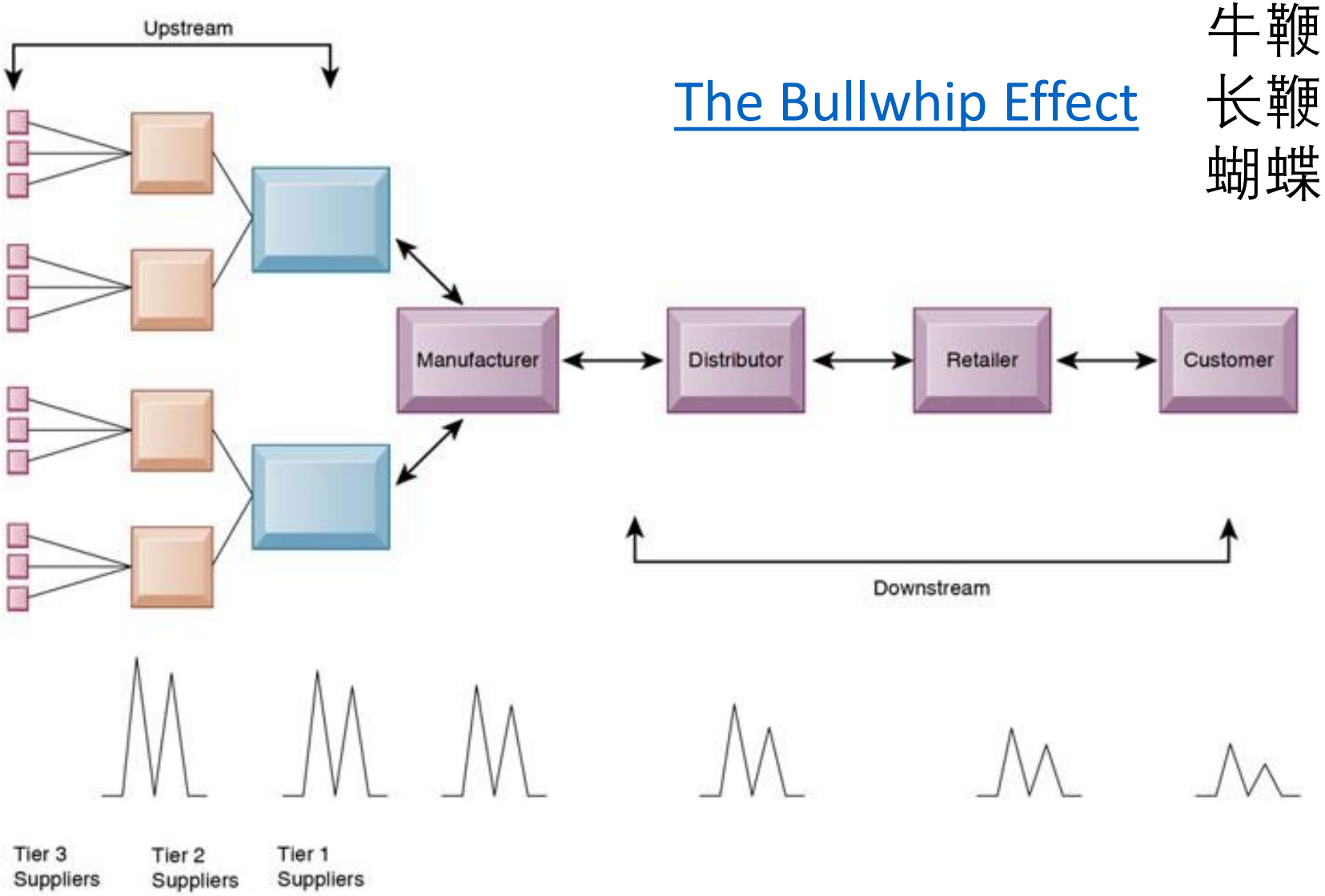
SCM Systems – Information in Supply Chain

- Inefficiencies
  - Parts shortages, underused plant capacity, excessive finished goods inventory, or high transportation costs
  - Caused by inaccurate or untimely information
- Uncertainty
  - uncertain product demand, late shipments from suppliers, defective parts or raw materials, or production process breakdowns.
  - safety stock acts as a buffer for the lack of flexibility in the supply chain: expensive inventory vs. opportunity loss

SCM Systems – Information in Supply Chain

牛鞭效应
长鞭效应
蝴蝶效应

The Bullwhip Effect



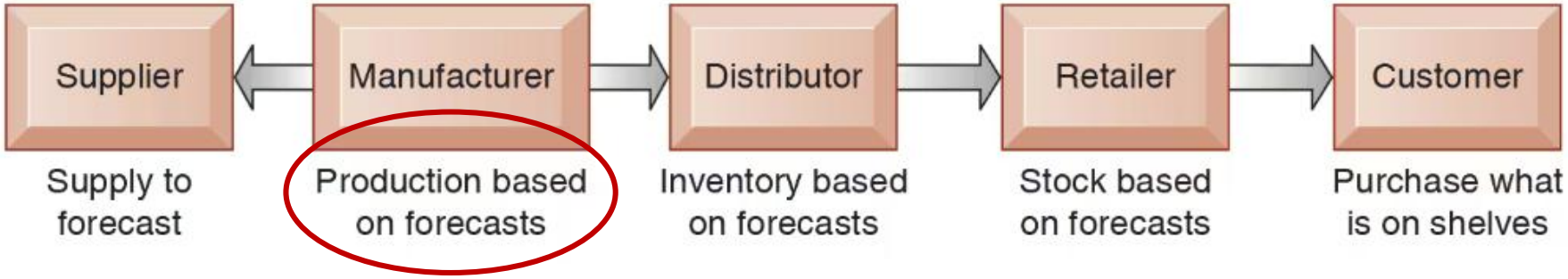
SCM Systems – SCM Software

- Supply chain planning systems
  - ✓Model existing supply chain
  - ✓Enable demand planning
  - ✓Optimize sourcing, manufacturing plans
  - ✓Establish inventory levels
  - ✓Identify transportation modes
- Supply chain execution systems
  - ✓Manage flow of products through distribution centers and warehouses

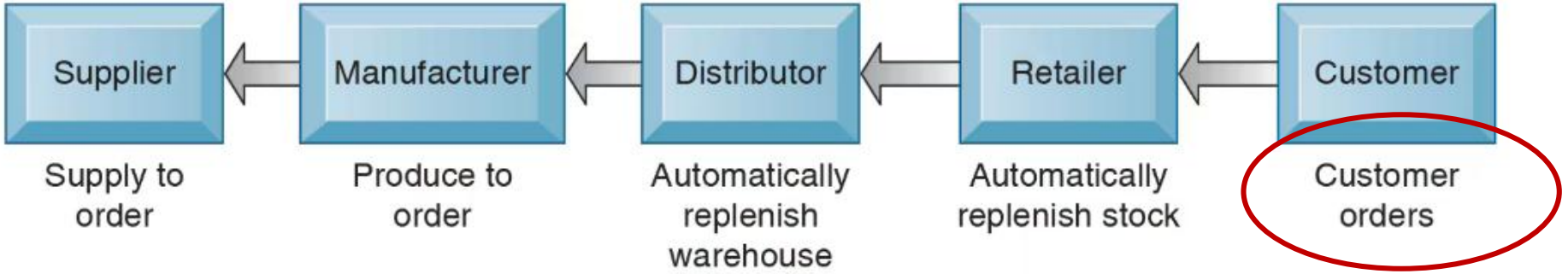
SCM Systems – Demand-Driven Supply Chains

Example?

Push-Based Model (推动式)

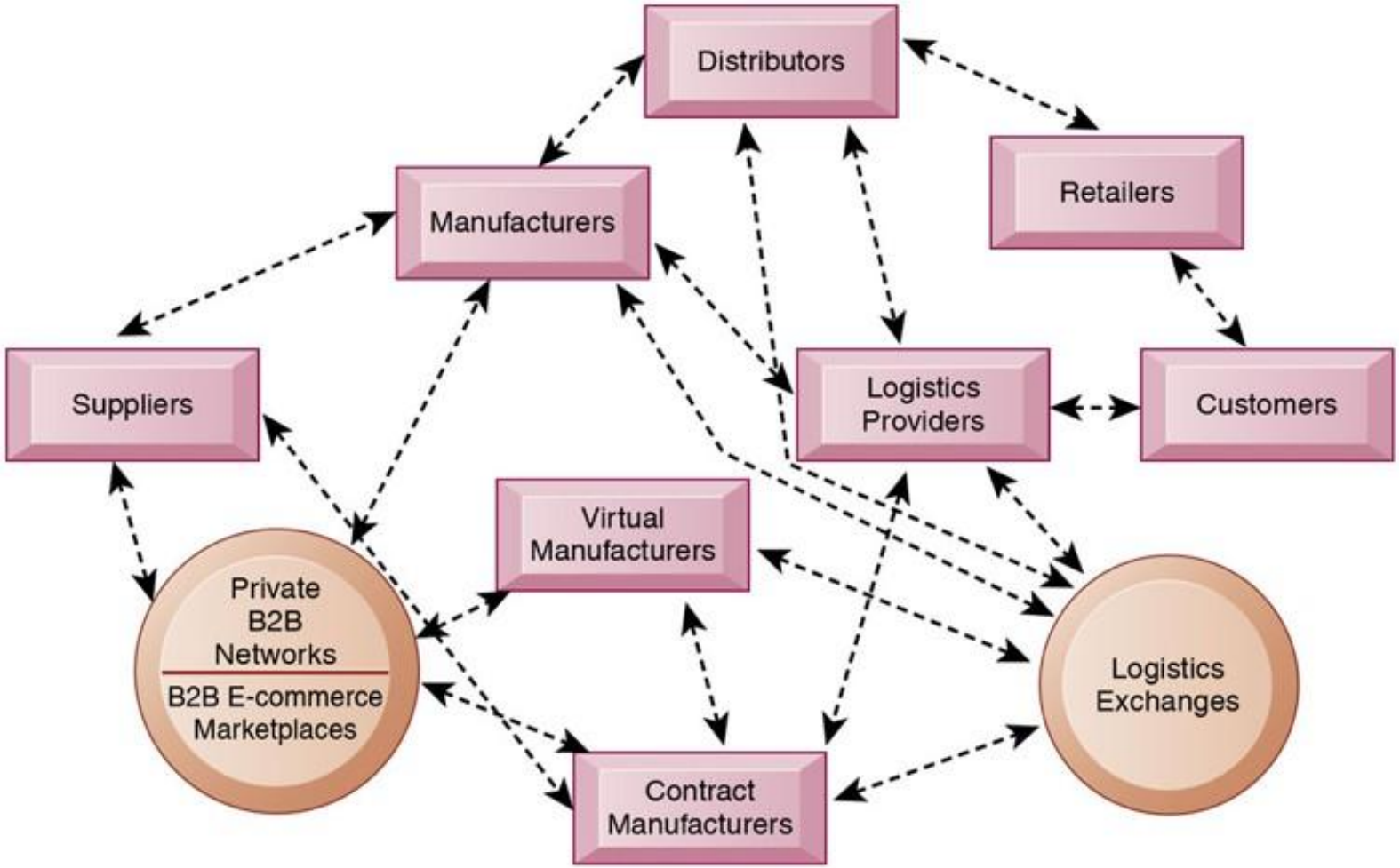


Pull-Based Model (拉动式)



SCM Systems – Internet-Driven Supply Chains

How should we interpret this web-like structure?



SCM Systems – Business Value

- Match supply to demand
- Reduce inventory levels
- Improve delivery service
- Speed product time to market
- Use assets more effectively
  - ✓Total supply chain costs can be 75 percent of operating budget
- Increase sales

SCM Systems – Global SCM Challenges

- Global supply chain issues
  - Geographical distances, time differences
  - Participants from different countries
    - Different performance standards
    - Different legal requirements
- Internet helps manage global complexities
  - Warehouse management
  - Transportation management
  - Logistics
  - Outsourcing

COVID-19

tariff war

Singapore 30 by 30 → food supply chain disruption

SCM Systems – AI Applications

- More accurate inventory management

  AI can manage inventory levels with greater precision and efficiency. AI-powered forecasting systems can use inventory information to forecast customer demand and adjust product stocking accordingly. **Computer vision systems** (such as cameras installed on supply chain infrastructure, vehicles, and drones) can calculate inventory levels in real time throughout the supply chain and monitor warehouse storage capacity.

- More timely deliveries

  Logistics companies use machine learning to train models that optimize and manage delivery routes as components move along the supply chain. These models can prioritize shipments based on order volumes, delivery promises, deadlines, customer importance, or product availability. AI can also provide all parties in distribution networks with more-accurate estimated times of arrival.

SCM Systems – AI Applications

- Improved supply chain sustainability

  AI can drive operational efficiencies that make supply chains more sustainable, such as reducing vehicle fuel consumption or wasted materials. More precise demand forecasts and production planning help ensure manufacturers don’t overproduce.

- More precise demand planning

  AI predicts demand using sales patterns, marketing leads, economic outlook, and market trends. AI-infused systems can estimate the impact of events like weather, downturns, or disruptions.

- Simulations for supply chain optimization

  AI-powered simulations visualize complex logistics networks and identify improvements. They help assess redesign impacts, optimize routes, and test alternative supplier or warehouse strategies.

SCM Systems – Quiz

Which of the following is not a true statement about the challenges of managing a global supply chain?

A. Global supply chains typically have participants from a smaller number of countries than domestic supply chains.

B. Global supply chains typically span greater geographic distances and time differences than domestic supply chains do.

C. Government regulations impacting supply chain management may vary from region to region.

D. Performance standards may vary from region to region or from nation to nation.

CRM Systems – Definition

Customer relationship management (CRM) system

- **Captures** and **integrates** customer data from all over the organization
- **Consolidates** and **analyzes** customer data
- **Distributes** customer **information** to various systems and customer touch points across enterprise
- Provides a single enterprise view of customers

What is CRM and How Does it Work? (by Salesforce)

CRM Systems – Definition

CRM Systems – Software Packages

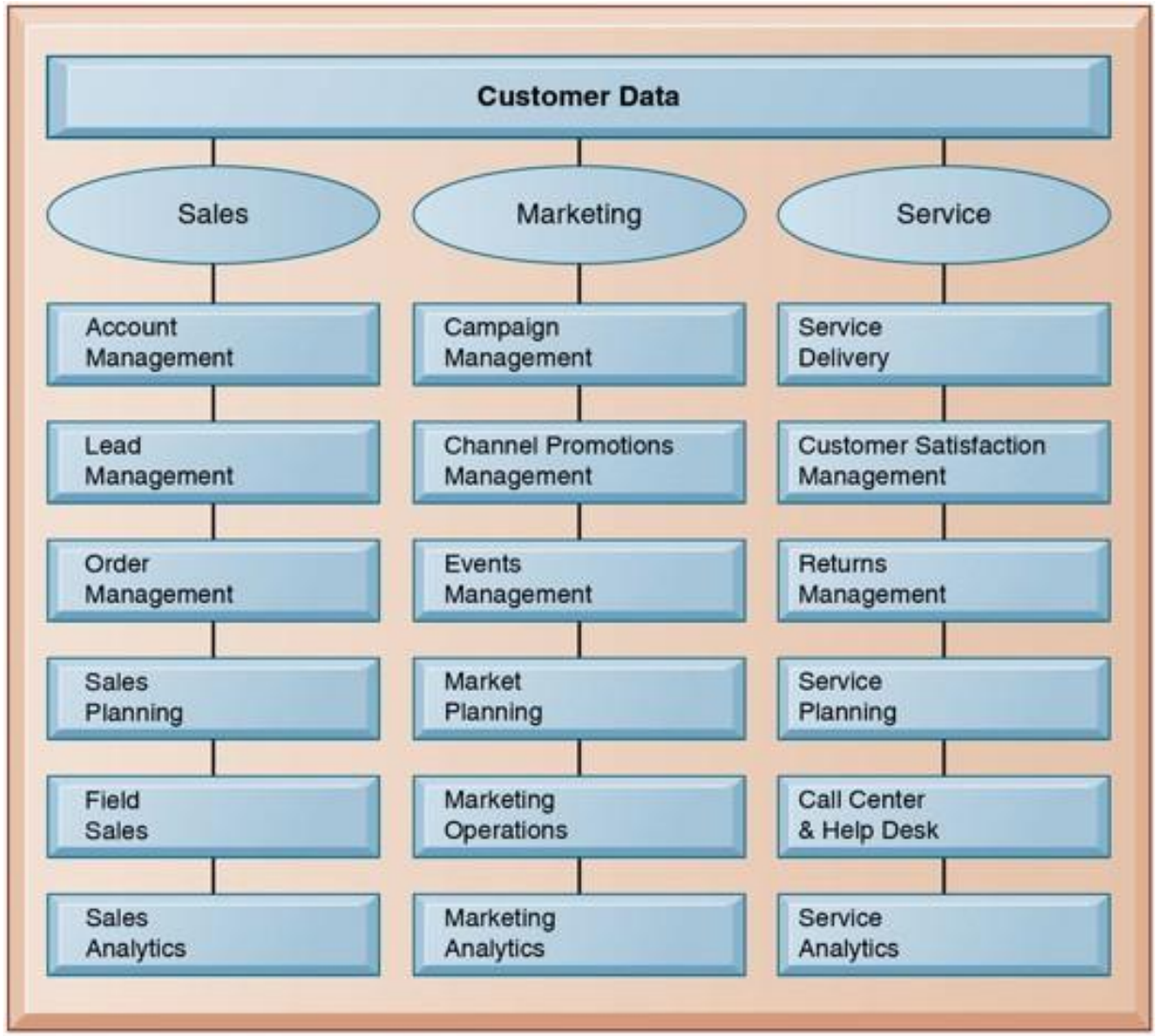
Commercial CRM system software packages range from niche tools to large-scale enterprise applications

- Sales force automation (SFA modules)
  - Sales prospect and contact information
  - Sales quote generation capabilities
- Customer service
  - Assigning and managing customer service requests
  - Web-based self-service capabilities
- Marketing
  - Capturing prospect and customer data, scheduling and tracking direct-marketing mailings or e-mail
  - Cross-selling

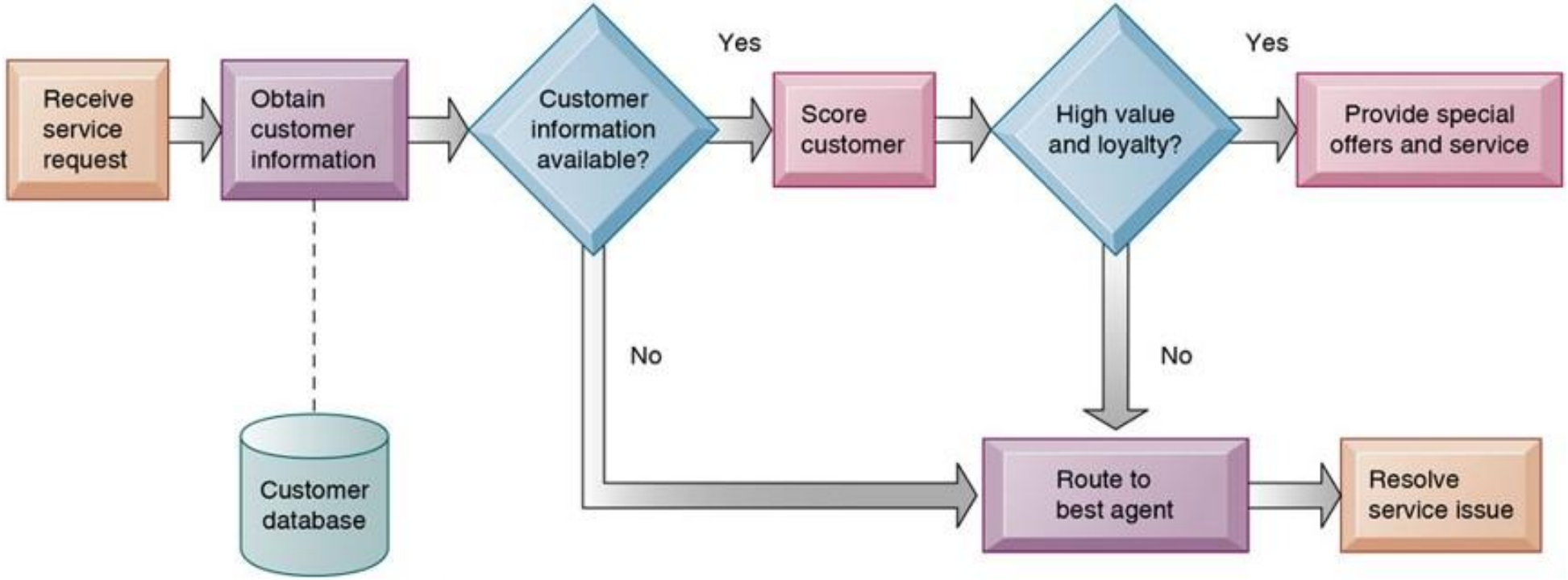
Oracle Sales Demo: Sales Automation and CRM Software

CRM Systems – Capabilities

Business processes supported by CRM systems.



CRM Systems – Capabilities



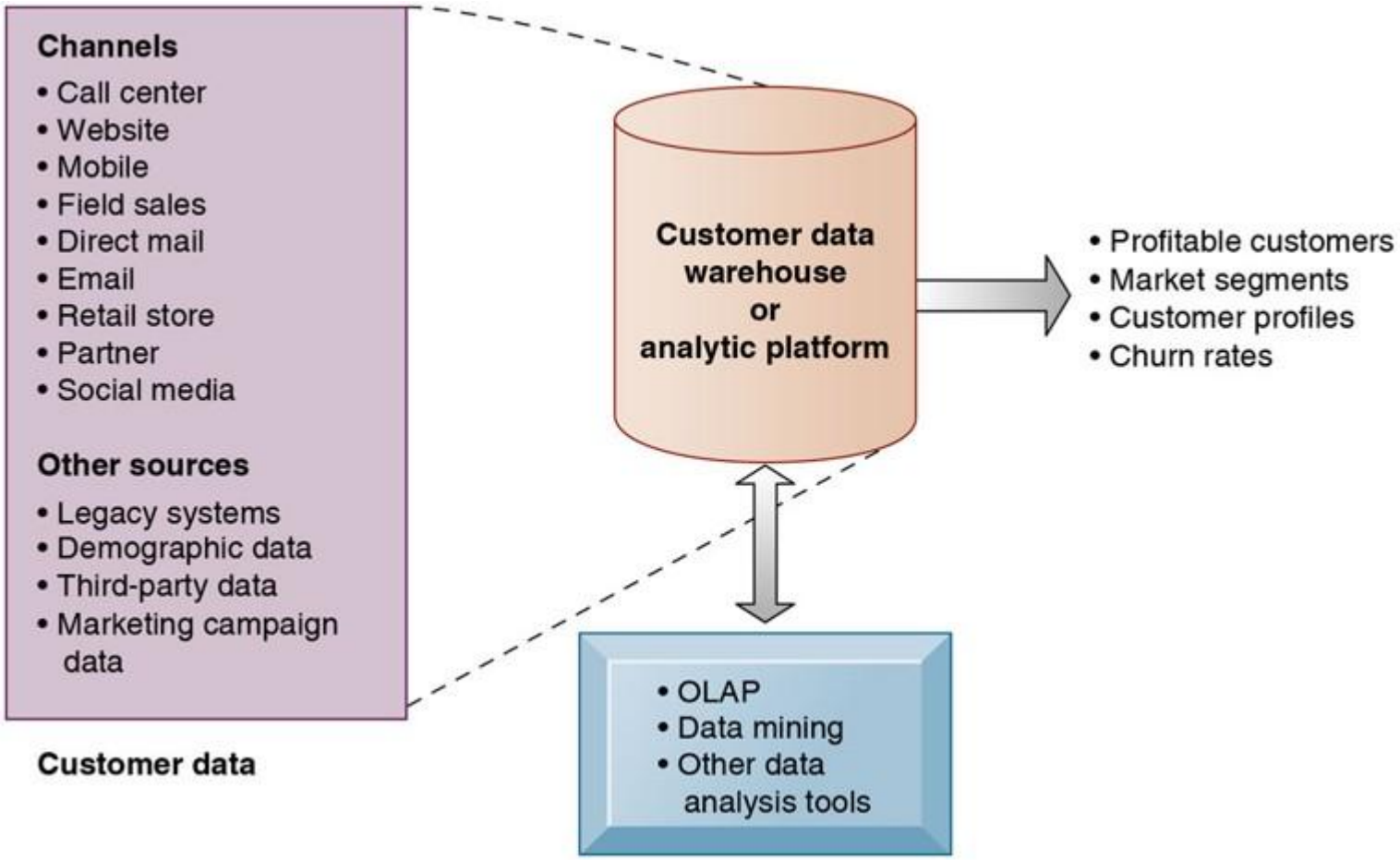
This process map shows how a best practice for promoting customer loyalty through customer service would be modeled by CRM software, helping firms identify high-value customers for preferential treatment.

CRM Systems – Types

- Operational CRM
  - Customer-facing applications
  - Examples: Sales force automation, call center and customer service support, marketing automation
- Analytical CRM
  - Applications that analyze the customer data generated by operational CRM applications
    - Examples: Online analytical processing (OLAP), data mining, customer lifetime value (CLTV)
  - To provide information for improving business performance

CRM Systems – Types

Analytical CRM



CRM Systems – Business Value

As a customer, how have you experienced a company's CRM system in action?

- Business value of CRM systems
  - Increased customer satisfaction
  - Reduced direct-marketing costs
  - More effective marketing
  - Lower costs for customer acquisition/retention
  - Increased sales revenue
- Churn rate
  - Number of customers who stop using or purchasing products or services from a company
  - Indicator of growth or decline of firm's customer base

CRM Systems – AI Applications

Capability	Description
Digital assistant agent	Use AI-powered conversational bots to respond to customer inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.
Field service routing optimization	Use predictive AI and machine learning with individual performance pattern data to optimize field service scheduling to maximize efficient use of resources and to help service teams complete more jobs per day.
Lead conversion and sales opportunity win probability	Provide sales reps with rapid identification of their best leads and sales opportunities with predictive scoring and win probabilities. Additionally, identify sales deals that are falling behind or need more attention.
Pricing optimization	Use AI-recommended pricing and discounts to strike a balance between achieving the most revenue with the highest likelihood to close.
Predictive scoring	Use machine learning to predict which leads and acts are most likely to engage and convert to buyers.

CRM Systems – Quiz

Analytical CRM systems are based on data from all of the following except:

A. CLTV
B. Operational CRM systems
C. Data from sources organized in data warehouses
D. Customer touch points

Enterprise Application Challenges

- High Cost & Complexity
  - ERP, SCM, CRM are costly and complex to implement
  - ~28% projects over budget; ~33% delayed (Panorama, 2024)
  - Causes: organizational issues, scope creep, customization
- Major Organizational Change
  - Requires process redesign and employee retraining
  - Resistance and poor data practices often cause failure
- Vendor & Data Dependence
  - High switching costs (SAP, Oracle)
  - Needs standardized data and cleansing
- Evolving Solutions
  - Simplified, cloud, and hybrid ERP models
  - Minimize customization to cut costs

AI Functionality Provided by Major Enterprise Application Vendors

Vendor	Functions	AI
SAP	ERP, CRM, SCM, HCM	Machine learning and generative AI integrated with SAP business processes and solutions. For example, SAP Business AI is built into SAP finance applications to manage capital effectively, improve reporting accuracy, mitigate fraud risk, and reduce losses with anomaly detection and prevention. Joule generative AI assistant acts as a work copilot across SAP business systems, answering questions with natural language.
Oracle	Oracle offers a range of AI solutions, including generative AI, AI services, and AI infrastructure.	AI services, including generative AI, have prebuilt machine learning models that make it easier for developers to apply AI to applications and business operations. The models can be custom trained. Oracle Cloud Infrastructure (OCI) Generative AI is a fully managed service to seamlessly integrate these versatile language models into a wide range of use cases, including writing assistance, summarization, analysis, and chat. Oracle Fusion Cloud CX includes generative AI-assisted answer generation, assisted scheduling for field service, opportunity quality scoring, and seller engagement recommendations.
Microsoft	CRM, ERP, primarily small to medium-size companies	Microsoft Dynamics 365 AI suite includes Dynamics 365 Copilot AI assistant across all business functions. Microsoft Dynamics Customer Insights, Microsoft Dynamics Sales Insights, and Microsoft Dynamics 365 Customer Service Insights are key offerings.
Salesforce	CRM	Einstein suite of AI tools includes Einstein GPT Generative AI for CRM, which delivers AI-created content across every sales, service, marketing, commerce, and IT interaction.

Summary

- **ERP** – Integrate core business processes
- **SCM** – Coordinate supply chains efficiently
- **Global SCM** – Manage cross-border complexity
- **CRM** – Strengthen customer relationships
- **Challenges** – High cost, change, and data issues
- **AI** – Drive automation, insight, and innovation